3AL Tutorial 11 2024 Solutions

1. Use the Third Sylow Theorem to prove that a group of order 77 is cyclic.

Let G be a group of order $77 = 7 \times 11$. By the Third Sylow Theorem, $n_7 \equiv 1 \bmod 7$ and $n_7 \mid 11$. Since the only divisors of 11 are 1 and 11, we must have $n_7 = 1$. We similarly have $n_{11} \equiv 1 \bmod 11$ and $n_{11} \mid 7$. Thus $n_{11} = 1$. This means we have a unique Sylow 7-subgroup of order 7. All elements of order 7 must be contained in this subgroup otherwise if there was some $g \in G$ with $|g| = 7$ and g not in the Sylow 7-subgroup we would have another Sylow 7-subgroup $\langle g \rangle$, which contradicts the uniqueness. Thus there are $7 - 1 = 6$ elements of order 7 (we take the total number of elements in the Sylow 7-subgroup and subtract 1, which accounts for the identity which has order $1 \neq 7$). Similarly, there must be $11 - 1 = 10$ elements of order 11 in G. In total, we have found $1 + 6 + 10 = 17$ elements. However, there are 77 elements in G. This means that there must be some elements of order other than 1, 7 and 11. By Lagrange's Theorem, these elements must have an order that divides 77. The only remaining option is that the order of these elements in 77. This means that for any of these elements of order 77, the cyclic group generated by it must equal G. So G is cyclic.

2. Let G be a group of order p^2q for distinct primes p and q with $p < q$. Prove that if G has more than one Sylow q-subgroup, then $|G| = 12$.

By the Third Sylow Theorem, we have $n_q \equiv 1 \bmod q$ and $n_q \mid p^2$. Since $n_q \neq 1$, we must have $n_q = tq + 1$ for some integer $t \geq 1$. Since $n_q \mid p^2$, we must have $n_q = p$ or $n_q = p^2$.

Suppose that $n_q = p$. Then $t = (p-1)/q$. However, $p < q$ by assumption, so this means $p = 1$. This contradicts p being a prime. Therefore, we must have have $n_q = p^2$. Then $t = (p^2 - 1)/q$. So $q \mid (p^2 - 1)$, meaning $q \mid (p+1)$ or $q \mid (p-1)$. We can discard the latter since $p - 1 < q$ and so it would force $p = 1$, a contradiction. Thus $q \mid (p+1)$. This means that $p = 2$ and $q = 3$. So we can conclude that

$|G| = 2^2 \times 3 = 12.$

3. Let G be a group of order p^2q^2 for distinct primes p and q with $q \nmid (p^2-1)$ and $p \nmid (q^2-1)$.

   (a) Let P and Q be normal subgroups of G with orders p^2 and q^2, respectively.

      (i) Show that $P \cap Q = \{1\}$.

      (ii) Prove that $G = PQ$.

      (iii) Prove that for $x \in P$ and $y \in Q$, we have $xy = yx$.

(i) Suppose $z \in P \cap Q$. Then $|z|$ divides both p^2 and q^2, which are distinct primes. The only possibility is that $|z| = 1$. That is, $z = 1$

(ii) Since P and Q are normal subgroups, PQ is a normal subgroup of G (Lemma 2.18). Moreover, since $P \cap Q = \{1\}$, the order of PQ is $p^2q^2 = |G|$. Thus $PQ = G$.

(iii) Let $x \in P$ and $y \in Q$. We claim that $xy = yx$, that is, $xyx^{-1}y^{-1} = 1$. Notice that because P and Q are normal, $xyx^{-1}y^{-1} \in xP = P$ and $xyx^{-1}y^{-1} \in Qy^{-1} = Q$. Thus $xyx^{-1}y^{-1} \in P \cap Q = \{1\}$. This means that $xyx^{-1}y^{-1} = 1$ and $xy = yx$, as desired.

   (b) Prove that G is abelian.

   Hint: First find subgroups P and Q that satisfy the conditions of (a).

By the Third Sylow Theorem, we have $n_p \equiv 1 \bmod p$. So $n_p = tp+1$ for some $t \geq 0$. We also have $n_p \mid q^2$. So, either $n_p = 1$, q or q^2. If $n_p = q$, we would have $q = tp+1$ and $p \mid (q-1)$, which contradicts $p \nmid (q^2-1)$. Similarly, if $n_p = q^2$, we would have $p \mid (q^2-1)$, another contradiction. Thus $n_p = 1$. Through similar reasoning, $n_q = 1$. Thus there are unique Sylow p- and q-subgroups P and Q, respectively. As they are unique, they are also normal. By (a), $G = PQ$.

Take $g, g' \in G$. Then we can write $g = xy$ and $g' = x'y'$ for some $x, x' \in P$ and $y, y' \in Q$. Then $gg' = (xy)(x'y')$. However, by (a) (iii), $yx' = x'y$ and so $gg' = xx'yy'$. Recall that a group of order p^2 (similarly, q^2) is abelian. Thus $xx' = x'x$ and $yy' = y'y$. So $gg' = x'xy'y = x'y'xy = g'g$. So, G is abelian.

4. Let G be a group of order $70 = 2 \times 5 \times 7$. Show that G has a normal subgroup of index 2.

   HINT: Find subgroups P and Q such that $|G : PQ| = 2$.

By the Third Sylow Theorem, $n_5 \equiv 1 \bmod 5$ and $n_5 \mid 14$. The only option is $n_5 = 1$, so there is a unique Sylow 5-subgroup of G, say P. Similarly, $n_7 \equiv 1 \bmod 7$ and $n_7 \mid 10$. Again, the only option is $n_7 = 1$, so there is a unique Sylow 7-subgroup, say Q. Since P is a unique Sylow 5-subgroup it is normal. By the Second Isomorphism Theorem, PQ is then a subgroup of G. Moreover, since $P \cap Q = \{1\}$, the order of PQ is $5 \times 7 = 35$. It then follows from Lagrange's Theorem that $|G : PQ| = 2$ and so PQ is a normal subgroup of G.

5. Prove that there is no simple group of order $825 = 3 \times 5^2 \times 11$.

Let G be a group of order 825. By the Third Sylow Theorem, $n_{11} \equiv 1 \bmod 11$ and $n_{11} \mid 75$. The divisors of 75 are $1, 3, 5, 15, 25$ and 75. Thus n_{11} must be 1. That is, there is a unique Sylow 11-subgroup of G. So we have a normal subgroup of order 11 that is nontrivial and proper. Thus, G is not simple.

6. Let R be a ring.

   (a) Prove that the identity of R is unique.

   (b) Prove that if an element $r \in R$ has a multiplicative inverse, then it is unique.

(c) Prove that $(-1_R)(-1_R) = 1_R$.

(a) Suppose 1_R and $1'_R$ are elements in R such that $1_R r = r 1_R = r$ and $1'_R r = r 1'_R = r$ for all $r \in R$. Then $1_R 1'_R = 1_R$ and $1_R 1'_R = 1'_R$ and so $1_R = 1'_R$.

(b) Let $r \in R$. Suppose there exists $r_1, r_2 \in R$ such that $rr_1 = r_1 r = 1_R$ and $rr_2 = r_2 r = 1_R$. Then $r_2(rr_1) = r_2 1_R = r_2$ and $(r_2 r)r_1 = 1_R r_1 = r_1$. So $r_1 = r_2$.

(c) We have $(-1_R)(-1_R) = 1_R 1_R = 1_R$ using Theorem 5.3 (iv).

7. Let R be a ring and $n \geq 2$. Suppose that $r^n = r$ for all $r \in R$. Let $a, b \in R$. Prove that $ab = 0$ implies $ba = 0$.

We have

$$ba = (ba)^n = (ba)(ba)(ab)^{n-2} = b(ab)a(ba)^{n-2} = 0.$$

8. The **characteristic** of a ring R is the smallest *positive* integer n such that $n1_R = 0_R$. If such a positive integer does not exist, the characteristic of R is said to be 0. The characteristic of R is denoted by $\text{char}(R)$.

(a) Let $m \geq 2$. Determine $\text{char}(\mathbb{Z}_m)$.

(b) Let R be a ring with $\text{char}(R) = n$. What can we say about $\text{char}(M_2(R))$?

(c) Prove that if $n = \text{char}(R)$ then $nr = 0_R$ for all $r \in R$.

(a) We have $m[1] = [m] = [0]$. Note that for any $1 \leq n \leq m$, $n[1] = [n] \neq [0]$. Thus $\text{char}(\mathbb{Z}_m) = m$.

NOTE: It is not enough to show $m1_R = 0_R$, we need to show it is the *smallest* positive integer that satisfies this.

(b) The identity of $M_2(R)$ is $\begin{pmatrix} 1_R & 0_R \\ 0_R & 1_R \end{pmatrix}$ and the zero is $\begin{pmatrix} 0_R & 0_R \\ 0_R & 0_R \end{pmatrix}$. For $m \geq 1$, we have

$$m\begin{pmatrix} 1_R & 0_R \\ 0_R & 1_R \end{pmatrix} = \begin{pmatrix} m1_R & 0_R \\ 0_R & m1_R \end{pmatrix}.$$

The smallest positive integer m that will make this the zero matrix is the smallest positive integer m such that $m1_R = 0_R$. But $\operatorname{char}(R) = n$, so $\operatorname{char}(M_2(R)) = n$.

(c) We have $nr = n(1_R r) = (n1_R)r = 0_R r = 0_R$.

Extra problems

(I) Let G be a group with order $p^n m$ where $n \geq 1$ and p is a prime such that $p \nmid m$. Suppose that H is a normal subgroup of G such that p divides the order of H. Prove that every Sylow p-subgroup of G/H can be written as PH/H for some Sylow p-subgroup P of G.

HINT: Think about orders, Correspondence Theorem, Lagrange's Theorem, First Sylow Theorem, Second Isomorphism Theorem.

Let $|H| = p^k d$ for some $1 \leq k \leq n$ and $p \nmid d$. We then have $|G/H| = p^{n-k}(m/d)$. Suppose $\mathcal{K}$ is a Sylow p-subgroup of G/H. Then $|\mathcal{K}| = p^{n-k}$. By the Correspondence Theorem, we can write $\mathcal{K} = K/H$ for some subgroup K of G such that $H \subseteq K$. Then, by Lagrange's Theorem,

$$|K| = |K/H||H| = |\mathcal{K}||H| = p^{n-k}p^k d = p^n d.$$

By the First Sylow Theorem, K has some Sylow p-subgroup P of order p^n. However, this is also a subgroup of G and so it P is also a Sylow p-subgroup of G. Note that $H \subseteq PH$, so we can consider PH/H. It remains to show that $\mathcal{K} = K/H = PH/H$. Since $H \subseteq K$ and $P \subseteq K$, we must have $PH \subseteq K$. Since H is normal in G, we can use the Second Isomorphism Theorem to get

$$|PH| = |P||H|/|P \cap H| = p^n p^k d/|P \cap H|.$$

We want to show that $|PH| = |K| = p^{n-k}$. Since $|P| = p^n$ and $|H| = p^k d$ for some $k \leq n$, we must have $|P \cap H| \leq p^k$. Thus

$$|PH| \geq p^n p^k d/p^k = p^n d \geq p^{n-k} = |K|.$$

Since $PH \subseteq K$, we can conclude that $|PH| = |K|$ and so $PH = KH$. Finally, $\mathcal{K} = PH/H$.